

CAVE Challenges and Use Cases: A Case Study of the X-SITE CAVE

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Figure 1: Interactive exploration of scientific datasets in the X-SITE CAVE using optical tracking and a handheld control device.

ABSTRACT

The X-SITE CAVE is a custom, high-resolution visualization platform that has a history spanning more than 17 years. In this paper we summarize the key challenges and considerations we faced over the past years and discuss advantages and disadvantages of design choices. Ongoing development efforts focus on accelerating the provision of new content and improving maintainability, while simultaneously lowering costs and staffing needs. Third-party open-source software is central to these efforts and is complemented by in-house developments, e.g., for customized calibration tools and system control software as well as an adapter to the Godot game engine. We also report successful use cases that highlight the benefits of large-scale VR environments in several application domains.

Index Terms: Virtual reality, visualization systems and tools, immersive displays, tiled displays, CAVE, game engines, scientific visualization.

1 INTRODUCTION

CAVEs are highly specialized large VR displays for visualization, simulation, and experimentation. They are also complex hardware-software systems with often unique designs constrained by client demands and operator budgets. Today, a wide variety of CAVE systems exists. In this paper we report useful principles we learned while maintaining and continuously modernizing the X-SITE CAVE. During that journey, we encountered typical challenges, such as a varying number of scientists and staff in the lab and uncertainty about future financial resources.

Once installed, the hardware can run for many years with only occasional calibration and few component replacements. However,

in everyday use, the software can be a decisive factor. Our rules of thumb for the successful continuation of the visualization laboratory with a minimal staff requirement are:

1. Keep it simple – avoid unnecessary dependencies
2. Minimize reliance on closed-source components
3. Choose popular, actively maintained FOSS
4. Implement custom code only when essential
5. Design for resilience to OS and library updates
6. Use open, standard file formats & protocols
7. Streamline authoring workflows
8. Decouple the rendering engine from VR content

2 RELATED WORK

The relevance of CAVE systems has been a subject of debate [1]. CAVEs entail high acquisition costs, occupy a large dedicated space, and demand ongoing maintenance and staff support. However, previous position papers highlighted the continuing importance of CAVE systems, which aligns with our experience with the X-SITE CAVE. Advantages include improved embodiment, natural collaboration of co-located participants and shared audio experiences [15, 19]. We present use cases where the X-SITE CAVE has proven beneficial in subsection 3.2.

Some CAVEs, including the X-SITE CAVE, use tiled projection systems. Comprehensive work on the calibration of tiled projector based displays has been published by Aditi Majumder [9]. The calibration routines have been revisited in [10] with a discussion of SOTA hardware options. A potential trend in terms of display hardware is the transition from projectors to LED panels [5, 12].

In addition to commercial (closed source) solutions, there are various free and open-source software (FOSS) alternatives. This

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Figure 2: CAVEs offer an ideal platform to present a variety of scientific projects to high-school students (©TUBAF).

includes middleware (e.g. VRPN), complete collaboration frameworks like COVISE [2, 20], integration of 3D web platforms [13] and integrations of scientific tools such as VTK and Paraview [18, 21]. In terms of rendering frameworks, there is a trend toward using game engines for easier authoring of VR applications for various platforms [11]. Development for multiple platforms can be streamlined further by using open standards such as X3D [14, 15], OpenXR [6, 18] and WebXR [4]. Scientific visualization labs benefit in various aspects from widely adopted standards and open file formats [15]. Commercial (closed source) components can reduce maintenance workload but also cause problems in the long term for institutions with low or varying budgets [22].

3 RELEVANCE OF CAVE SYSTEMS

A full CAVE system typically costs more than a set of consumer VR headsets for a comparable audience size. But is it always possible to replace large-scale immersive displays with headsets? The advantages and disadvantages depend on the specific use case. We will highlight the potential benefits of CAVE-like systems with our local X-SITE CAVE as an example case. Nevertheless, many of these arguments apply to other systems as well.

Let's start with a thought experiment: we are planning a scientific presentation for a diverse audience. The participants come from various fields (scientists, companies, government agencies, general public visitors) and span a wide age range. This is a typical scenario for using the X-SITE CAVE as a visualization platform. Beyond internal scientific applications, our CAVE plays a key role in the university's public-relations initiatives (see Figure 2).

Each of the participants gets an HMD, a sufficiently scaled play boundary and has to get familiar with the hardware - which may require instructions by experts (onboarding). Because HMDs must be thoroughly cleaned for each subsequent user, a dedicated cleaning step is also required. During the event all participants experience the identical scene and have to explore the scientific content collaboratively. This entails continuously following the gaze and gestures of any participant or speaker while discussing the virtual worlds, presentation slides or videos. Furthermore, motion sickness has to be considered by allowing easy exit from the virtual space. The use case therefore demands well-built collaborative XR applications that support natural multi-user interaction [6, 19]. Additionally, space constraints and the need for a sufficient number of instructors can increase onboarding time and staff requirements.

In contrast to the described HMD scenario, entering and exiting a CAVE is straightforward [19]. If stereo vision is not used, no wearable devices are required at all. Thus, onboarding time can be eliminated and events for split audiences can be repeated with little effort. A single presenter is sufficient to start and run presentations in the X-SITE CAVE.

3.1 Limitations

Space and acquisition costs can be limiting at the beginning while especially public institutions may struggle even more with long-term operating expenses because funds are often available only for a limited period, and follow-up financing is not guaranteed. Especially maintenance by external service providers and subscription-based software licenses induce high running costs.

Proprietary software without source code access or continuous development introduces an additional problem: specific features for scientific applications might be missing and if software is no longer maintained or updated, support for newer applications and hardware may be compromised. Staff members who are specifically responsible for the development and maintenance of the visualization labs are often unaffordable for smaller institutions.

A key question is: where are needs for sophisticated setups and where is potential for simplification? Pros and cons of commercial solutions must be weighed against open-source or self-developed alternatives. Some applications may require a sophisticated custom audio setup, large-scale precise tracking or stereo vision to enhance immersion and interact with close objects. When choosing an audio system, conventional 5.1 and 7.1 surround sound setups are limited but provide broad software support out of the box. As an alternative to expensive professional tracking systems, Kinect cameras or comparable depth and tracking cameras are often readily available in academic VR laboratories. These cameras can be a sufficient solution for various use cases, as demonstrated by Gonçalves et al. with the €5.000 KAVE system [7, 8].

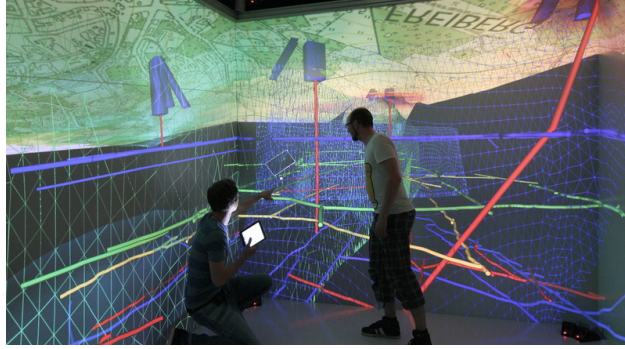
3.2 Use Cases

A key advantage of using CAVE systems for scientific presentations is the natural collaboration and interaction with visitors combined with fast and uncomplicated onboarding. Besides optional stereo glasses, no wearable devices are required, allowing users to enter and exit the projection space on demand. This facilitates multiple runs with large audiences to be carried out quickly one after the other. The main purpose of the X-SITE CAVE is to provide a visualization platform for scientific presentations. These presentations typically involve audiences of up to 15 participants who immerse themselves in 3D models, simulations, and point clouds from various fields. Many of the typical use cases of the X-SITE CAVE cannot be seamlessly replaced by HMD-based VR applications.

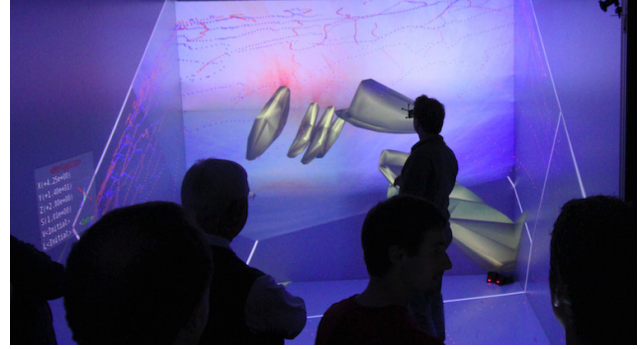
A majority of visualized datasets originate from the domain of geosciences, including geological subsurface models (Figure 3a), geophysical surveys (Figure 3b), and 3D scans of underground mines. Cooperative research projects in robotics and environmental sciences have contributed sonar scans that can be augmented with additional geodata for improved context. Game engines enable us to integrate animated replicas of autonomous, environment monitoring vessels to demonstrate data collection. These applications are also of great interest to authorities in the fields of environment, land surveying, and infrastructure planning, as well as to companies in the transportation sector. Consequently, the audience can be very heterogeneous in terms of specialist areas and prior VR experience.

The interactive visualization of HPC simulations is another important pillar of our applications. A CAVE environment proved to be ideal for inspecting both macroscopic and microscopic datasets in interdisciplinary teams. During a long-term collaborative project "Multifunctional filters for molten metal filtration" (Figure 1), the system was used regularly for project meetings (Figure 4a).

Besides geoscientific applications, the cultural heritage sector is an ideal source of detailed 3D environments that can be explored in group experiences. In addition to established 3D meshes and point clouds (Figure 4b), the visualization of Gaussian splatting models is a highly relevant topic today. Because Gaussian splats can reproduce the background and surroundings of scanned objects or buildings in an appealing fashion, the extensive display surface



(a) 3D subsurface model of local mine shafts



(b) Presentation of seismic surveys

Figure 3: Geo-scientific visualizations are a common use case that brings together a broad audience.

of a CAVE can be used effectively. Other applications include the CAVE as experimental lab to investigate benefits of VR for specific cases or to evaluate inside-out tracking quality using artificial features on the display wall [3]. Public relations has become another major use case for our CAVE, as it provides a convenient way to present a wide range of scientific projects to visitors at a single location during 60- to 120-minute events. School classes in particular enjoy visits to the CAVE, turning it into a central element of the university's student recruitment.

Over time, a pool of scenes with scientific content from various domains has accumulated. This also provides a constantly changing set of examples for general presentations in the X-SITE CAVE. Scientific topics with related simulations and 3D environments can be experienced immersively, combined with videos and images that explain the background and original data acquisition.

4 THE X-SITE CAVE

The X-SITE CAVE is a distributed system comprising 16 computers, handhelds for system control, and a highly tiled projector setup (25 projectors in total) for three walls and a floor projection. The left wall has been extended beyond the traditional cube layout to a 6 m x 2.6 m display. We described the past and current hardware setup in [17]. An important characteristic of our CAVE is the transition from specialized hardware and software to off-the-shelf consumer hardware and open-source software. Tracking is still provided through a commercial optical (infrared) tracking system. Integrated and running from the start, the tracking system has proven its worth and justifies the initial investment. Staff reductions forced us to simplify maintenance and accelerate content creation. Today the system can be maintained and operated by a single scientist in part-time. Outdated proprietary calibration software for the display walls caused incompatibilities with modern rendering engines, as the calibration tool was no longer maintained by the developer.

During latest hardware upgrades we decided to omit the stereo-3D feature of our initial projectors. While this compromise reduces depth perception and immersion, it drastically lowered cost and system complexity. The removal of this feature had no significant impact on the typical usage scenarios (see subsection 3.2) as stereo-3D is primarily beneficial for a single person and can increase motion sickness for other viewers who are not being tracked. Furthermore, the stereo quality of the original projectors was insufficient for the tiled display arrangement. Even though we gave up stereoscopic projection, head- and body-tracking remains available on-demand.

From the early start we rely entirely on Linux operating systems, providing us robustness, flexibility and reliability. We chose a distribution that offers a rolling release. Community-developed open interfaces / middleware such as VRPN take a lot of work off our

Table 1: Overview of hardware and software currently in use.

Projectors	12+6+6+1 (left, front, right, floor)
Computing	1 workstation + 13 clients
Operating system	Manjaro Linux
Audio	7.1 surround sound
Calibration	In-house / open-source (C++, Python)
Interaction	IR optical tracking (commercial)
Middleware	VRPN (open-source)
Rendering engines	Godot, Paraview, Unreal Engine, OSG

hands. Regardless of the rendering engine, it has proven useful for us to have a general main application that handles all CAVE-related functionality and to load the actual content on-demand during run-time. This approach enables fast scene changes during presentations and facilitates easier authoring of new content. In addition, updates to the used rendering framework are less likely to conflict with existing content. We previously worked with an OpenScene-Graph (OSG) based framework, but now we primarily rely on the open-source game engine Godot (see subsection 4.1). Both frameworks use the described on-demand scene switching.

The glTF 3D file format with its wide adoption and standardized, open approach solved many issues which inhibited the exchange of 3D data between different modeling, rendering, and domain specific software products. Today importers and exporters for glTF are ubiquitous and enable seamless integration of 3D content.

4.1 GodotTD

The Godot tiled display framework is designed for distributed systems that follow a client-server paradigm. The number of clients is scalable, but the client and server applications can run on the same computer for small setups. It is in regular use since 2023 and is now our main rendering engine for the X-SITE CAVE. We released a stable version in 2024 under open-source license [16].

Godot allowed us to streamline the authoring of VR content considerably while offering easy coding and maintenance of the code base. The scenes created and exported with Godot proved to be robust to version changes, allowing older content to be loaded in newer releases of the framework without issues. The code required only minimal or no changes with minor Godot upgrades, and the Godot documentation provides sufficient guidance.

The approach of loading individual scenes into a single main application allows researchers and students to contribute content without needing in-depth knowledge of CAVES. If no specific interaction paradigms are required, arbitrary Godot projects can directly be exported as a resource pack, uploaded to central storage, and



(a) Analysis of fluid and particle simulations



(b) Precise point cloud of a historic building

Figure 4: CAVEs provide a platform for collaborative inspection of 3D scans and computational fluid-dynamic results.

displayed in the CAVE. For scenes which are just composed of existing 3D models (e.g. 3D scans, exported glTF files from scientific software), a simple scene can be assembled and distributed in just a few minutes – even on computers without sophisticated graphics hardware and without having to worry about CAVE specialties.

Although glTF files facilitate the easy integration of textured 3D meshes, integrating specific scientific datasets and their visuals still demands manual effort and domain-specific conversion (e.g. flow fields, lines, grids, particles). Currently, the integration is performed with Paraview, Blender and automated Python scripts. Recent additions to our Godot applications focus on directly integrating point cloud and Gaussian splatting datasets.

4.2 CAVE tools

Our current development work is centred on streamlining everyday CAVE operation and simplifying maintenance. We commit to using FOSS and standards wherever possible and develop additional tools with open-source libraries and a small codebase. In 2019 we replaced our legacy calibration software with an in-house tool based on the work of Aditi Majumder [9]. Recent changes affect communication in our distributed system and improve system control.

The distributed nature of the X-SITE system requires a flexible and reliable network protocol. Several legacy components still existed that used custom network commands to start and stop applications or to communicate with handheld devices during presentations. Our latest system control tool is based on the standardized and widely used *MQTT* protocol. For this publisher-subscriber approach, a *Mosquitto* (open-source) broker is running on the server node (workstation). All clients, including the workstation, periodically publish status information and subscribe to control message topics. MQTT allows us to replace custom components with community developed FOSS and to have a consistent communication channel. MQTT also supports encryption and permission management as additional security layer.

Most of our supplementary CAVE tools are programmed in Python, as we favor rapid development, maintenance and integration of existing libraries over higher performance – which is rarely critical for dashboards and helper tools. A Miniconda environment guarantees robustness against library updates. Our modular, lightweight Python dashboard visualizes client status, performance metrics, and application states. Projectors, computers, and tracking tools can be toggled by operators (see Figure 5). Applications such as rendering engines can be started remotely on all clients. The client’s monitoring instance continuously verifies the correct execution of the launched application and forces a restart if necessary.

A multimedia module enables image, audio, and video playback on any wall to support existing VR scenes across applications. Authorized MQTT users can subscribe to topics related to scene

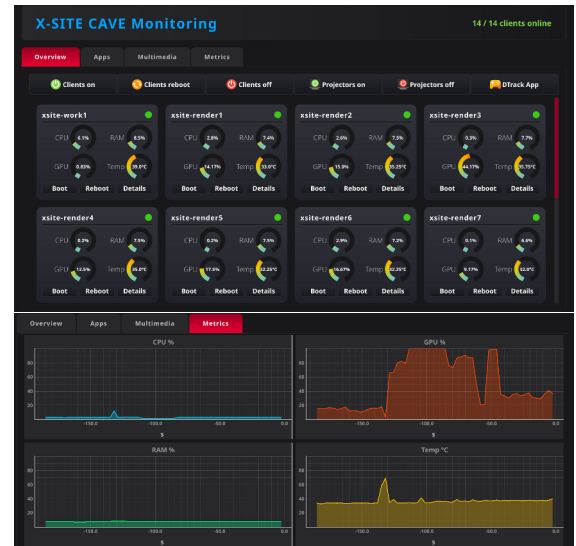


Figure 5: The new dashboard monitors system metrics of the distributed clients and provides easy access to essential functions.

changes, settings, or triggered application events, while any number of handheld devices can be authorized to act as control devices. The scene control and configuration are provided as a responsive website for maximum platform independence.

5 CONCLUSION

A transition to off-the-shelf hardware and the decision to omit 3D stereo allowed us to dramatically reduce the costs of CAVE modernization. The revision of the related software is a great help to us in the long term. Game engines offer a convenient and straightforward path for content creation. Minimalist frameworks particularly limit maintenance and enable workload reductions. Therefore, we transitioned to our Godot-based framework as the primary rendering engine. Supplementary tools for system control and monitoring rely on FOSS, open standards, and in-house developments.

We described our typical use cases to highlight the persistent relevance and benefits of CAVE-like immersive displays. In our experience, COTS headsets fail to meet our requirements for natural, collaborative data exploration during scientific presentations and project meetings. A valuable byproduct is the availability of numerous ready-to-use VR scenes that can be showcased for public relations with minimal onboarding effort.

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